Now we can look at the Point Slope form of a linear equation.

Slope intercept form
y=ax+b

Point slope form
y-y1=a(x-x1)

Standard form
Ax+By=C

Slope formula
a= (y-y1)/(x-x1)

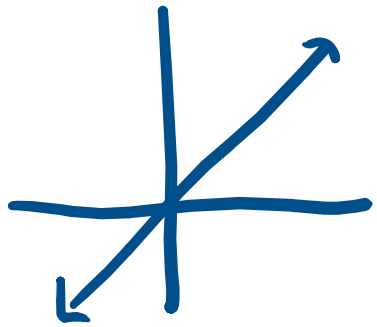
$$y = mx + b$$

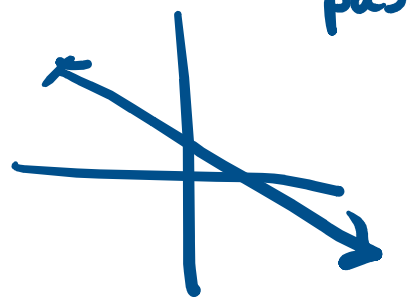
$$y - y_1 = m(x - x_1)$$

$$Ax + By = C$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point Slope form of a linear equation.
Notice that I said linear equation rather than function as this is not in function format. It is still a function but written as an equation in two variables with y as a function of x.





pass vertical line test
✓functions

Point slope form
y-y1=a(x-x1)

Reminder:
Not all equations in two variables are functions. Vertical lines where x equals a constant are not functions.

∅ function

Graphing a line from a point and a slope

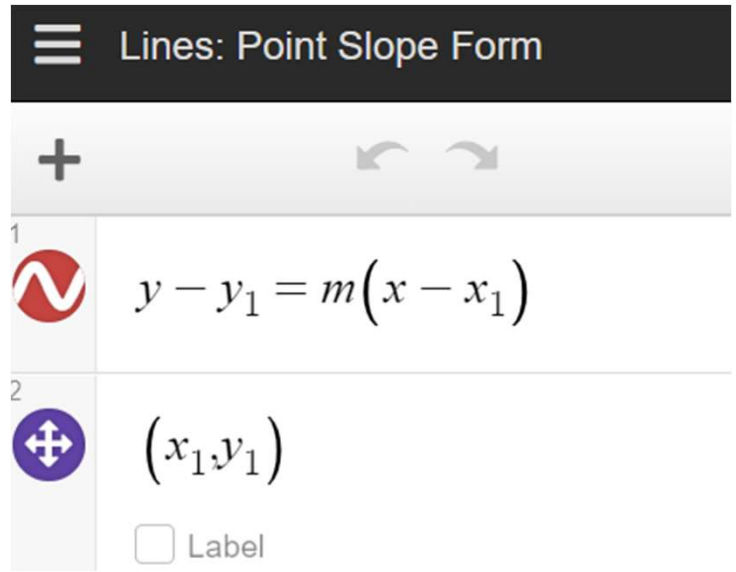
From the point (1,3) and a slope of 2, you can go up two every time you increase x by 1 to get to another point on the line.

In Desmos, go to the three bars

On the left and pull down

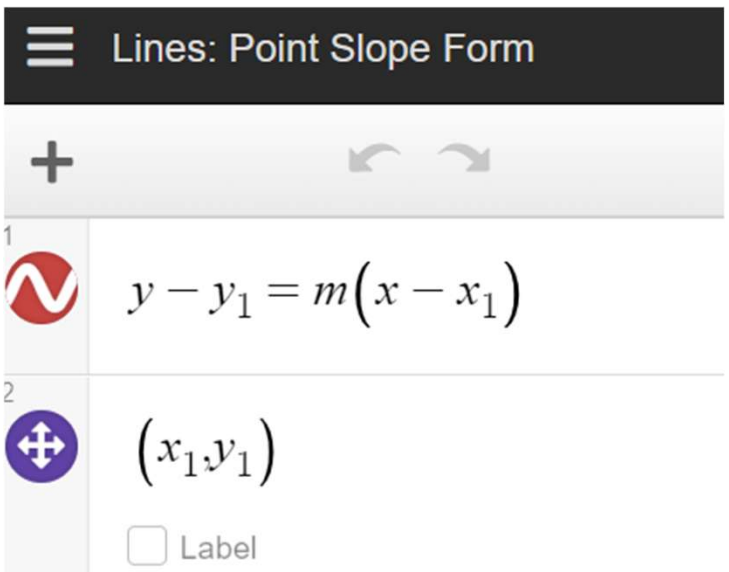
Lines: Point Slope Form

Type in y1, x1, and m



Graphing a line from a point and a slope

From the point (1,3) and a slope of 2, you can go up two every time you increase x by 1 to get to another point on the line.



$$m = \frac{(y - y_1)}{(x - x_1)}$$

m slope

(x_1, y_1) point

(x, y) any other point

$$m(x - x_1) = y - y_1$$

$$y - y_1 = m(x - x_1)$$

From a point and a slope, we can graph and find b by looking or algebraically.

When we know one point and the slope of the line, we can find b.

mx+b=y y=2x+b

b=-mx+y

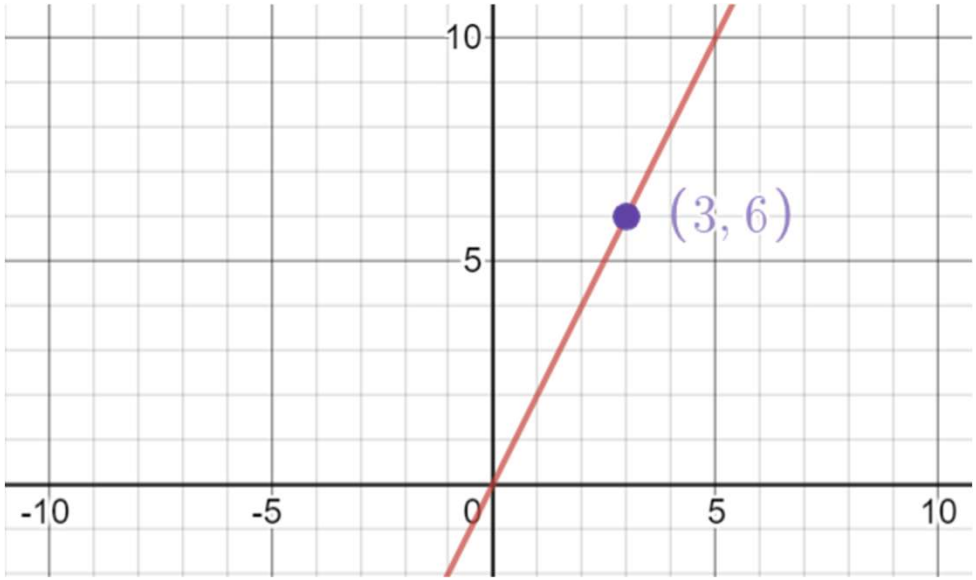
b=y-mx

(3,6) and a slope of 2

(x,y) m

6=(2)3+b

b= 6-2(3)



You can rearrange the slope formula for an equation of a line.

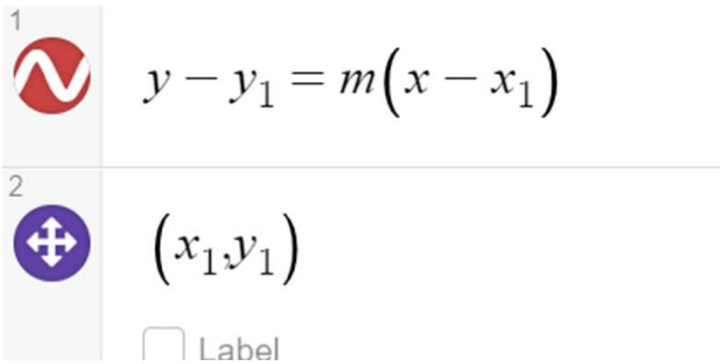
$$m=\frac{y-y1}{x-x1}$$

m(x-x1)= y-y1

y-y1=m(x-x1)

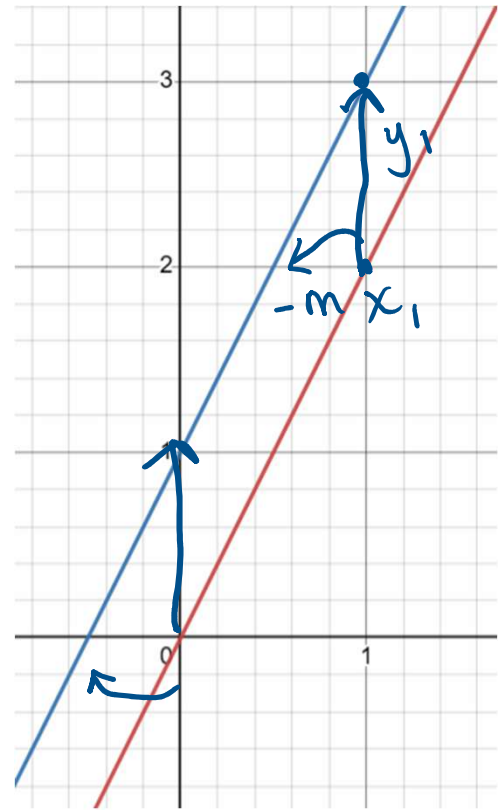
This is called the point slope form of a linear equation.

y=a(x-x1)+y1



y-y1=m(x-x1)

y=mx + (–mx1+y1)
b=y1-mx1
If we have one point and a tilt, then we have to figure out the other point that will satisfy the equation when x is zero and that is the y-intercept.

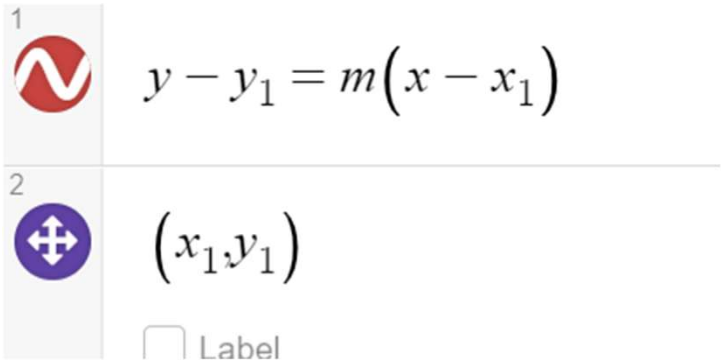


$y = 2x + 1$
$(1, 3)$

$y = 2x + 0$
$(1, 2)$

y_1

$$m = \frac{y - y1}{x - x1}$$



m(x-x1)= y-y1

y-y1=m(x-x1)

y=mx + (–mx1+y1)

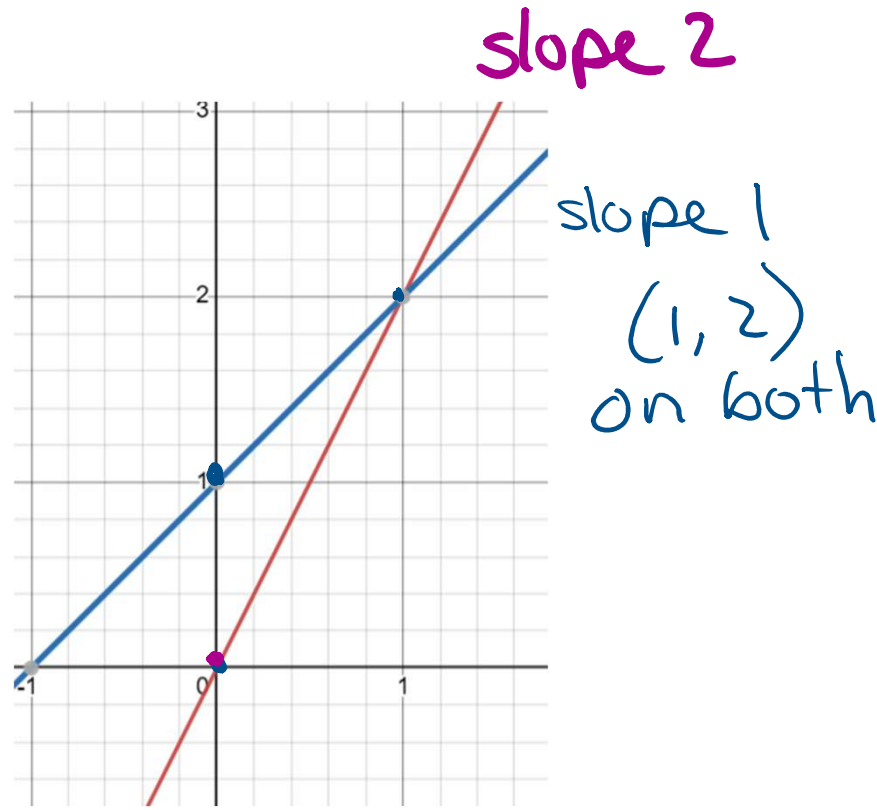
This is called the point slope form of a linear equation.

y=mx + (–mx1+y1)

b=y1-mx1

y=mx + (–mx1+y1)

The y-intercept depends on the slope and the other point that we choose.



y=mx + (–mx1+y1)

0 if x_1 is 0

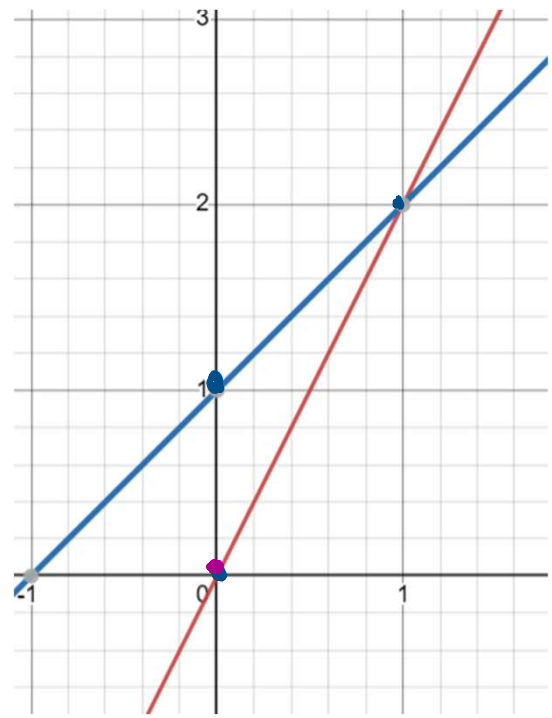
Notice that if we are given the y-intercept at the point, then it simplifies to the point slope form.

$(0, y_0)$

$y = mx + -m0 + y_0$

$y = mx + y_0$

slope 2



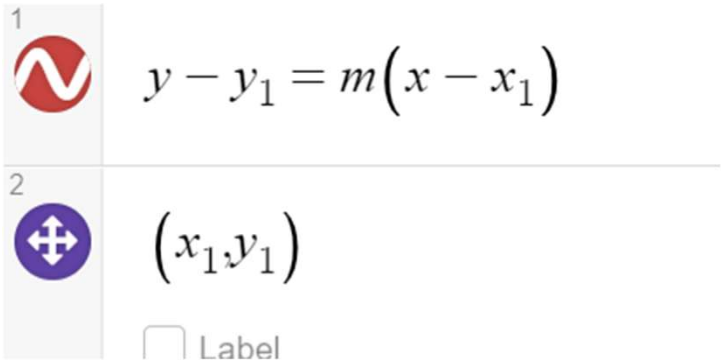
slope 1

(1, 2) on both

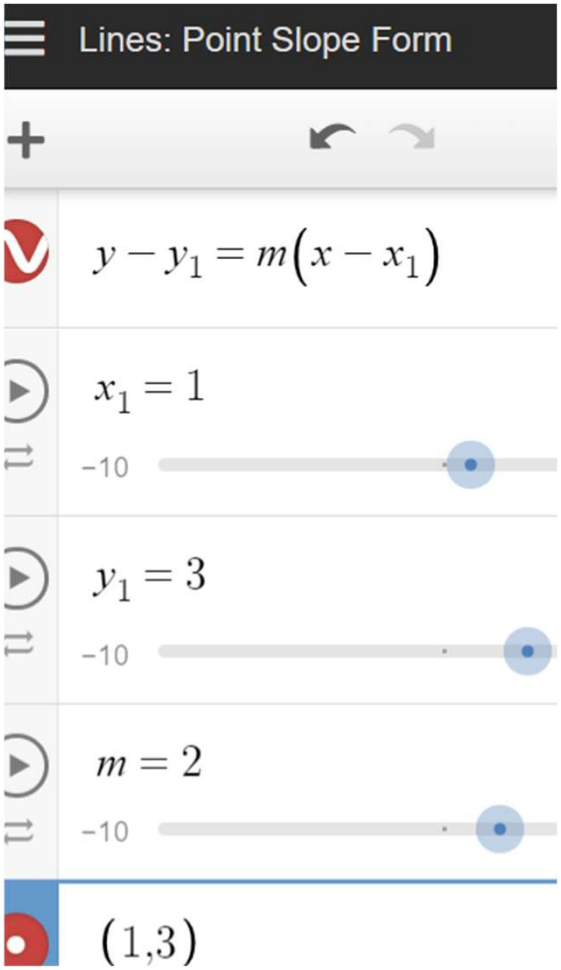
y-y1=m(x-x1)

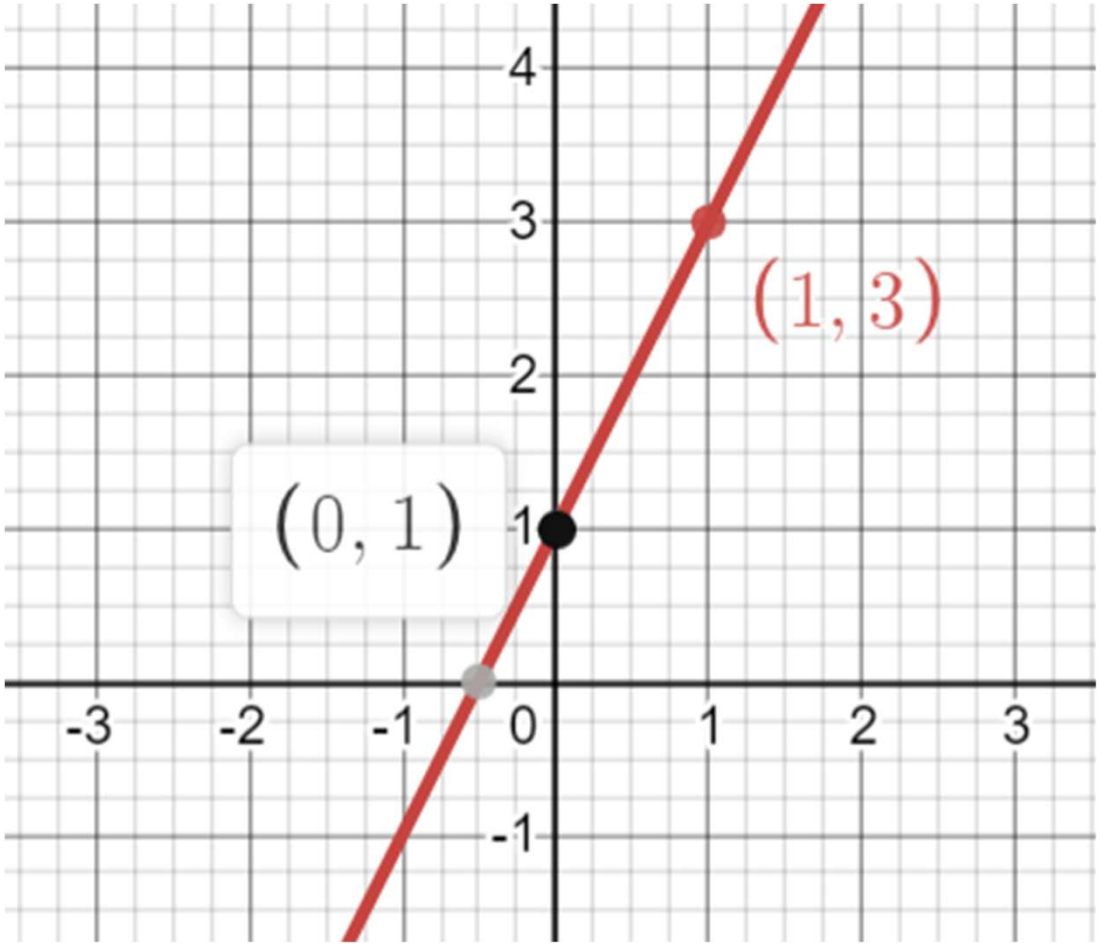
y=mx + (–mx1+y1)

b=y1-mx1



With point slope form in Desmos





$y-y1=m(x-x1)$ point slope form

(3,6) slope 2 $(-mx1+y1)=b$

$-2(3)+6=0$

$y-6=2(x-3)$

$y-6=2x-6$

$y=2x$

(2,5) slope 7

$y-y1=m(x-x1)$

$y-5=7(x-2)$

$y-5=7x-14$

$y=7x-14+5$

$y=7x-9$

Point slope form to Slope intercept form

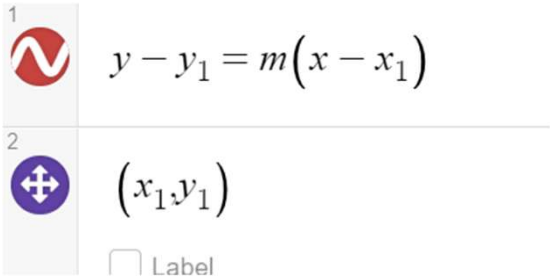
When point is (3,6) and slope is 2

y-6=2(x-3)

y-6=2x-6

y=2x-6+6

y=2x

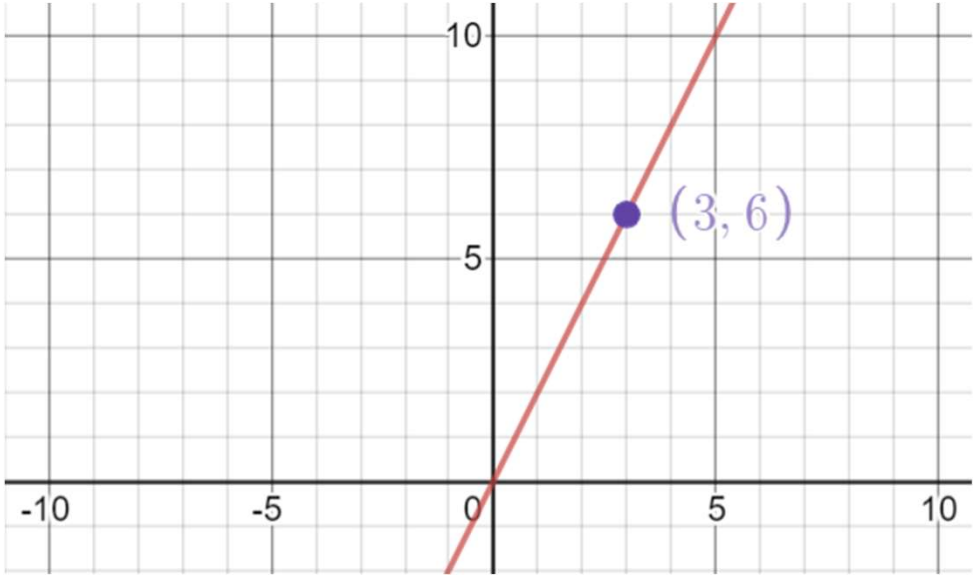


Point, slope form

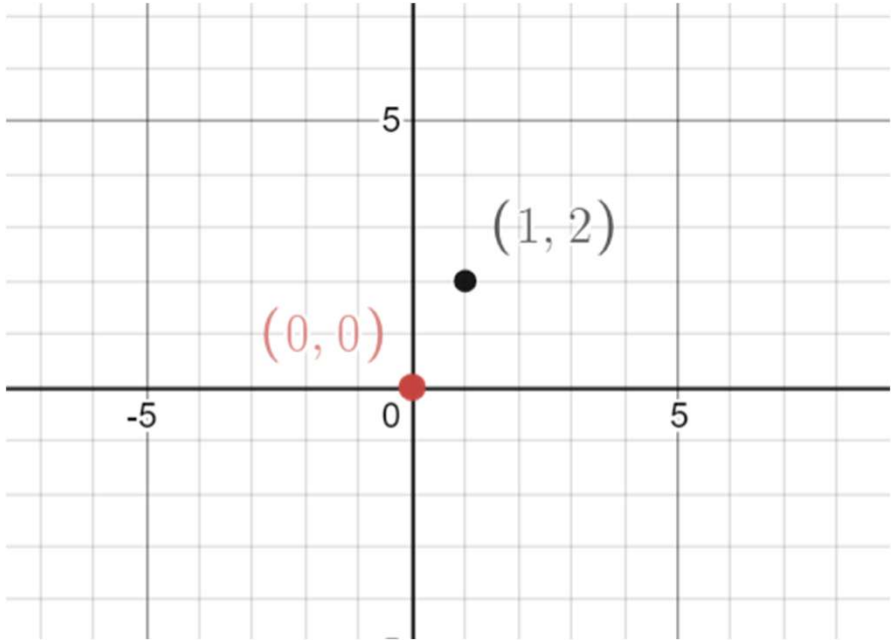
y-y1=m(x-x1)

Slope intercept form

y=mx+b



Two points can define a line. What if you have two points and want to get the y=ax+b form? (0,0) and (1,2)



Slope of a line is the change of y over change of x

Rise over run

slope or rate

$$\frac{\Delta y}{\Delta x} \quad \frac{\text{change of } y}{\text{change of } x}$$

$$m = \frac{y2 - y}{x2 - x}$$

If there are two points, it is the y of the first minus the y of the second over the x of the first minus the x of the second.

$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

It doesn't matter which is first or second, but you have to be consistent. (3,5) and (1,2)

slope: $\frac{5-2}{3-1} = \frac{3}{2}$ $\frac{2-5}{1-3} = \frac{-3}{-2} = \frac{3}{2} = 1.5$

$\frac{5-2}{3-1}$

Input two points: (x1,y1) and (x2,y2)

The slope= $\frac{y2-y}{x2-x}$ or $\frac{y1-y}{x1-x2}$

The key is that it is change of y over change of x and one point minus the other. (y2-y1)/(x2-x1)=

Finding the slope then using the point slope formula and one point

(1,3) and (2,5)

Slope $\frac{5-3}{2-1}$ = 2

Use (1,3)

y=ax+b

3=2(1)+b

3-2=b

b=1

y=2x+1

(2,5) and slope of 2 find b

x,y m

y-5=2(x-2)

y=2x-4+5

y=2x+1

To program a computer

Input two points, (x1, y1) and (x2, y2)

The equation is $y=(\frac{y2-y}{x2-x})\ x + y1 - (\frac{y2-y1}{x2-x1})\ x1$

Because the slope is $\frac{y2-y}{x2-x}$

and b is $b=y1 - mx1$

$b=y2 - mx2$

Slope intercept form

$$y = mx + b$$

Point slope form
(x1,y1) slope m

$$y - y_1 = m(x - x_1)$$

Standard form

$$Ax + By = C$$

Slope formula
Slope m
(x1,y1) (x2,y2)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$